

Decision Tree

1. **Setup Environment**
2. **Load Dataset**
 - a. Select a dataset (any suitable **regression** dataset).
 - b. Understand the features and target variable.
3. **Preprocess Data**
 - a. Handle missing values if any.
 - b. Normalize or standardize features if required.
 - c. Split dataset into training and testing sets.
4. **Initialize Decision Tree Classifier**
 - a. Choose appropriate parameters (criterion = "gini").
 - b. Set maximum depth and other hyperparameters as needed.
5. **Train the Model**
 - a. Fit the Decision Tree on the training dataset.
6. **Make Predictions**
 - a. Use the trained model to predict outcomes on the test dataset.
7. **Evaluate Model Performance**
 - a. Calculate RMSE, R2 Score, MAPE, MDA, FB, EVS etc.(any three)
8. **Visualize the Tree**
 - a. Plot the decision tree structure to interpret the splits.
9. **Experiment with Parameters**
 - a. Change tree depth, splitting criterion, and minimum samples per leaf.
 - b. Compare model performance under different settings.
10. **Summarize Results**
 - a. Report accuracy and metrics for different parameter choices.
 - b. Highlight which features were most important in classification.